

ML methods

Mathematics of AI - RPTU Kaiserslautern

WiSe 25/26

Topics of the lecture

- ML methods
 - Visualizing decision boundaries
 - Hyperparameter tuning
- Interpretability
 - Permutation importance

Decision Trees & Random forests

In the `0_decision_trees_sklearn.ipynb` notebook, you can:

- apply decision tree to the classification and regression problems.
- visualize the tree using `plot_tree`.

Similar notebooks are provided for random forests and SVM.

- try changing the dataset (to `load_iris`)
- explore the effect of different hyperparameter selections

ML methods

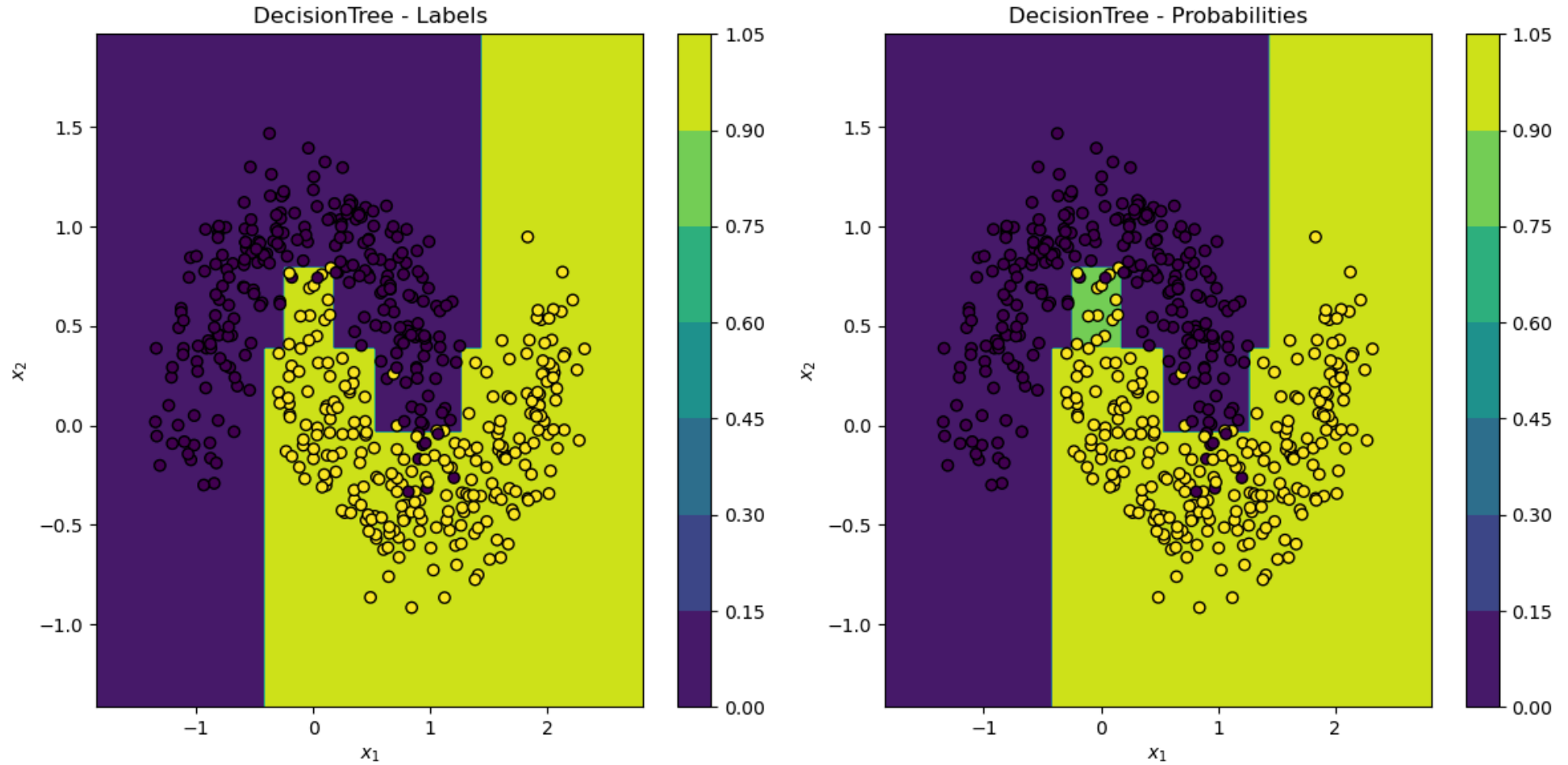
In the first notebook, we will use some of the sklearn classification methods, and visualize their decision boundary.

- use make moons dataset
- generate a grid
- create a function to:
 - fit an arbitrary model to the dataset
 - evaluate the model (i) on the dataset and (2) on the grid using `predict` and `predict_proba`
 - scatter the points and use `contourf` for the grid results.

The methods

You can use any method, the recommended ones to try are:

- `sklearn.linear_model.LogisticRegression`
- `sklearn.svm.SVC` with `kernel="linear"`
- `sklearn.svm.SVC` with `kernel="poly"` (change degree and `coef0`)
- `sklearn.svm.SVC` with `kernel="rbf"` (change `gamma`)
 - ▶ **note: to have `predict_proba` with `SVC` use `probability=True`**
- `sklearn.neural_network.MLPClassifier`
- `sklearn.tree.DecisionTreeClassifier`
- `sklearn.ensemble.RandomForestClassifier` (`n_estimators`)



Permutation importance

Use the permutation importance function to calculate feature importances in a model agnostic way. Try it on models that are trained on the same dataset and compare the results.

```
imps = permutation_importance(  
    clf,          # A trained classifier  
    X, y,         # A data set  
    n_repeats = 50, # Number of samples  
    random_state = 42 # Random seed for reproducibility  
)
```